基

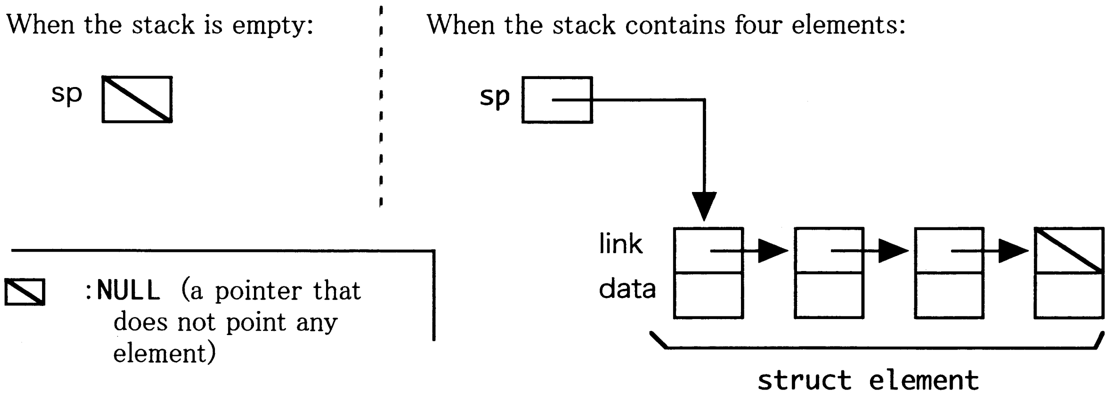
平成１９年度 筑波大学大学院 博士前期課程 システム情報工学研究科
コンピュータサイエンス専攻 入学試験問題

きそかだい すうがく じょうほうきそ すべ もんだい かいとう
基礎課題（数学と情報基礎の全ての問題に解答すること）

じょうほう きそ 情報基礎	以下の問題 I、問題 II を解答せよ。各解答は別々の答案用紙に記入すること。 **Solve both Problem I and Problem II, describing in two different sheets for them.**

Problem I

We consider a stack that is manipulated with two functions push() and pop(). The function **push()** is the function that adds an element to a stack. The function **pop()** removes an element from a stack. Although the function **pop()** returns an error when the stack is empty, we assume that no error occurs in this problem. We realize a stack that can contain integers by using a linked list. The following figures show such a stack in the boxes and arrows notation.



(1) The following is a program that adds an element to a stack written in the C language. Fill the boxes and complete the program. Note that `malloc()` is a function that allocates a memory block of a specified size (in bytes) and returns its address. In this problem, assume that the memory allocation always ends successfully. The constant `NULL` means a pointer that does not point any element.

```
#include <stdlib.h>

struct element {
    struct element *link;
    int data;
};

struct element *sp;
```

```
void push( int data ) {
    struct element *e;
    e = malloc( sizeof(struct element) );
    e->link = [    (A)    ];
    e->data = [    (B)    ];
    [    (C)    ];
}
```

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Continues from the previous page.

(2) In the following program, the function `sub()` removes two integers from the stack, and adds their difference to the stack. The function `mul()` removes two integers from the stack, and adds their product to the stack. The function `print()` removes an integer from the stack, and prints its value to the display.

```
void sub(void) {
    int arg1, arg2, res;
      arg2 = pop();
      arg1 = pop();
      res = arg1 - arg2;
      push( res );
}
```

```
void mul(void) {
    int arg1, arg2, res;
      arg2 = pop();
      arg1 = pop();
      res = arg1 * arg2;
      push( res );
}
```

```
void print(void) {
    int data;
    data = pop();
    printf("%d\n",data );
}
```

When the following function `main()` is executed, what is printed to the display?

```
main( int argc, char *argv[] ) {
   push( 10 );    push( 20 );    push( 30 );
   mul();         sub();         print();
}
```

(3) The following is a program that exchanges the top two elements of the stack written in the C language. Fill the boxes and complete the program.

```
void swap(void) {
    int arg1, arg2;
    arg2 = pop();
    arg1 = pop();
    [    (D)    ];
    [    (E)    ];
}
```

(4) The following is a program that reverses the sign of the top element in the stack written in the C language. Fill the boxes and complete the program.

```
void negate(void) {
    push( [    (F)    ] );
    swap();
    [    (G)    ];
}
```

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Problem II

Consider a directed graph which has n vertices $1, 2, \ldots, n$. Let us represent such a graph with n by n matrix whose (i, j) element is 1 if an edge from vertex i to j exists and 0 otherwise.

Graph G in fig.1 can be represented with 3 by 3 matrix shown in fig.2.

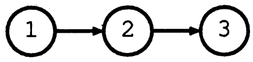


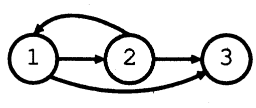
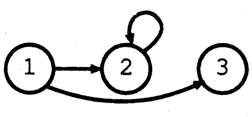
fig.1 : Graph G

$$\begin{array}{c c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \end{array}$$

fig.2: Matrix representing G.

Here we call a graph *reflexive* if, for any vertex i, there exists an edge from i to i itself. We call a graph *transitive* if, for any pair of edges from i to j and from j to k, there exists an edge from i to k. Finally, we say a graph is *symmetric* if, for any edge from vertex i to j, there exists an edge from j to i.

1. Following the example of fig.2 above, answer 3 by 3 matrices representing the graphs below.

(a)  (b) 

2. Following the example of fig.1, draw a reflexive and transitive graph with the least number of edges that has the same vertices as G in fig.1 and contains all the edges of G.

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3. Consider a matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ that represents a graph H of $n = 2$.

   (a) Using p and s, write a Boolean expression that becomes 1 if the graph H is reflexive, 0 otherwise.

   (b) Draw an active-high circuit that represents the Boolean expression obtained in (a) above, using two NAND gates. The circuit should take p and s as input and should generate output $z1$.

   (c) A Boolean expression that becomes 1 if the graph H is symmetric and 0 otherwise can be written as $q \cdot r + \overline{q} \cdot \overline{r}$. Using De Morgan's Theorem, convert the expression into an expression without disjunction (logical-OR).

   (d) Draw an active-high circuit that represents the Boolean expression obtained in (c) above, using five NAND gates. The circuit should take q and r as input and should generate output $z2$.